CSA06 CO3 — KartIQ Case Study

Consolidated Program Outputs

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CSA06 CO3 — KARTIQ CASE STUDY OUTPUTS
\nMODULE 1 — STABLE MERGE SORT
Input: [(105, 'O1'), (102, 'O2'), (105, 'O3'), (101, 'O4'), (102, 'O5'), (103, 'O6')]
Sorted: [(101, 'O4'), (102, 'O2'), (102, 'O5'), (103, 'O6'), (105, 'O1'), (105, 'O3')]
Comparisons: 9
Stable equal timestamp 102: ['O2', 'O5']
Stable equal timestamp 105: ['O1', 'O3']
For 10,000,000 orders, log2(n)= 24
Approximate n*log2(n) comparison scale: 230,000,000 (upper scale estimate)
\nMODULE 2 — RANDOMIZED THREE-WAY QUICK SORT
Input: [42, 17, 63, 8, 55, 29] Sorted: [8, 17, 29, 42, 55, 63]
Comparisons: 26 Partition sizes (less,equal,greater): [(5, 1, 0), (1, 1, 3), (0, 1, 2), (0, 1, 1)] Max recursion depth: 4
Nearly sorted 100,000 sellers: randomized pivot gives expected O(n log n);
n*log2(n) comparison scale: 1,600,000 (not an exact measured count)
\nMODULE 3 — FIRST-OCCURRENCE BINARY SEARCH
Timestamp: 101 first index: 0 order: (101, 'O4') comparisons: 4
Timestamp: 102 first index: 1 order: (102, 'O2') comparisons: 4
Timestamp: 105 first index: 4 order: (105, 'O1') comparisons: 4
Timestamp: 999 first index: -1 order: NOT FOUND comparisons: 2
Maximum lower-bound iterations for 10,000,000 items: about ceil(log2(n+1)) = 24
\nMODULE 4 — STRASSEN MATRIX MULTIPLICATION
A: [[1, 7], [3, 5]] B: [[1, 3], [7, 5]]
Conventional result: [[50, 38], [38, 34]] scalar multiplications: 8
Strassen result: [[50, 38], [38, 34]] scalar multiplications: 7
For n=1024 conventional multiplications: 1073741824
For n=1024 Strassen multiplications (7^10): 282475249
Ratio conventional / Strassen: 3.801
\nVALIDATION
All validation tests: PASS
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